

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)

Group No: 12

Team Members:

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1. Problem Title:

Contact Management System

STRUCTURAL DESIGN

2. Class diagram

(Draw class diagram that shows different classes and their relationships. Use UML notations)

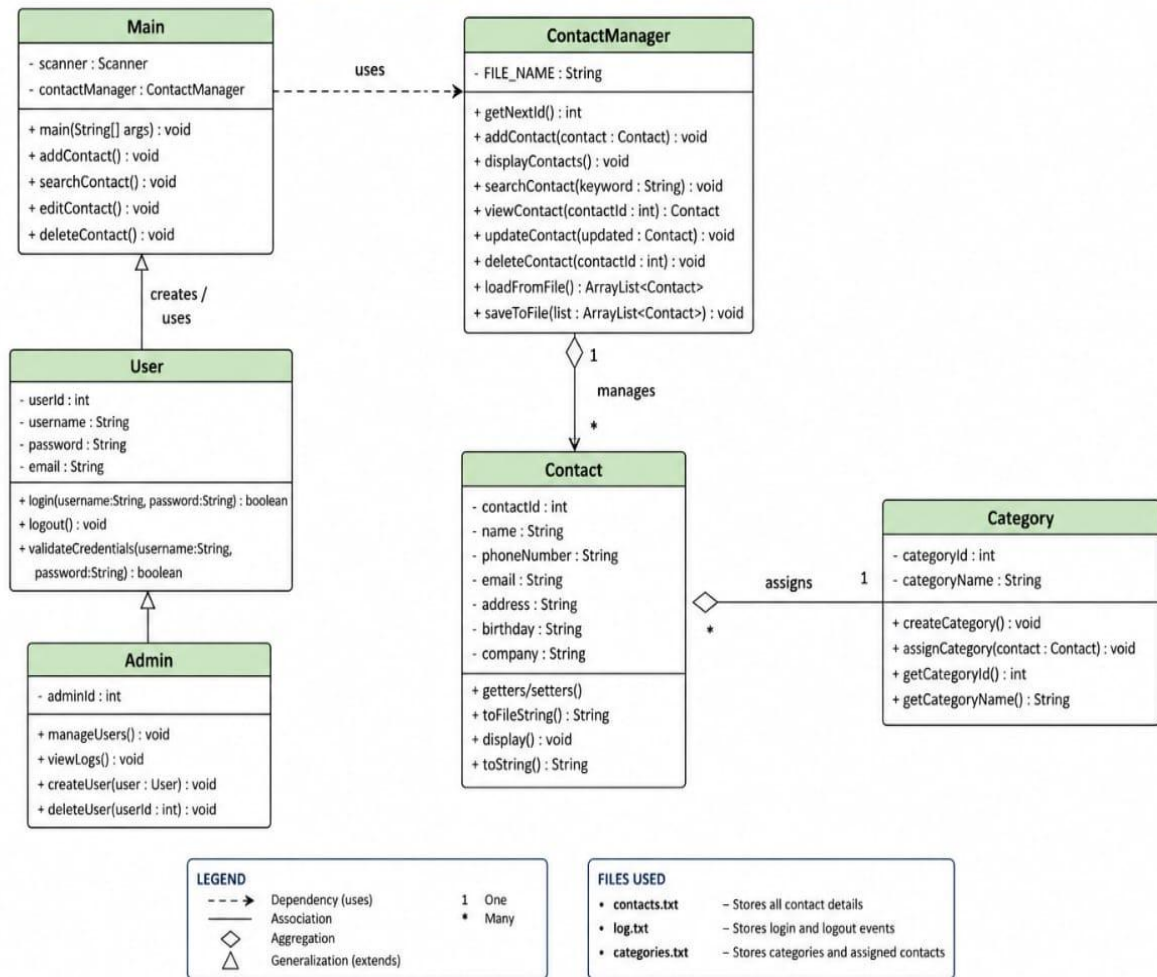
Classes:

- **User**
 - userId
 - username
 - password
 - email
- **Contact**
 - contactId
 - name
 - phoneNumber
 - email
 - address
 - birthday
 - company
- **ContactManager**
 - FILE_NAME (contacts.txt)
- **Category**
 - categoryId
 - categoryName

RELATIONSHIP BETWEEN CLASSES

From	Relationship	To / Description
User	Uses	ContactManager — User calls CRUD operations via menu
ContactManager	Manages	Contact — loads, stores, updates, and deletes Contact objects
Contact	Serialized to	contacts.txt — one contact per line in CSV format
Category	Groups	Contact — assigns contacts to named categories, saved to categories.txt
User	Logs to	log.txt — every login/logout recorded with timestamp

CLASS DIAGRAM – CONTACT MANAGEMENT SYSTEM



BEHAVIOURAL DESIGN

3. Use-case diagram

a. List of actors

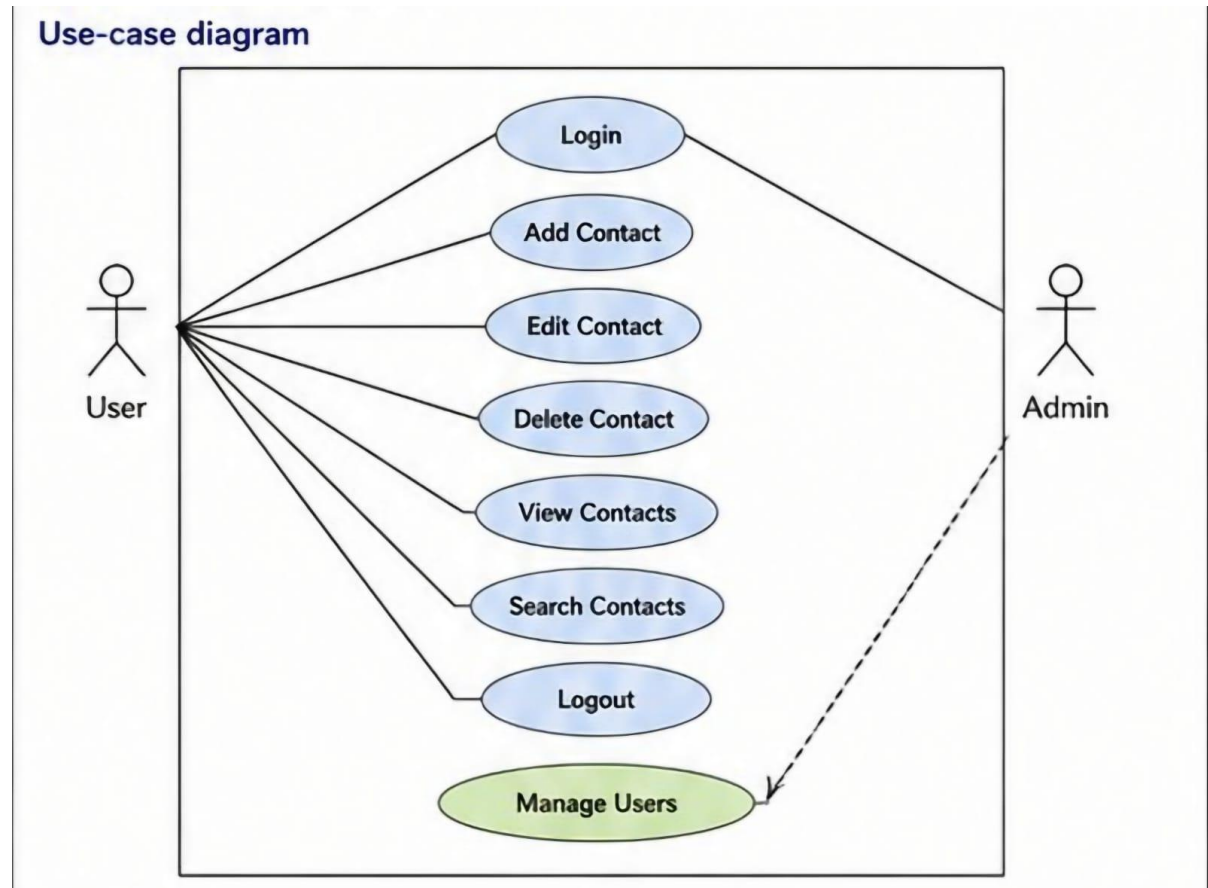
1. User – Manages contacts
2. Admin – Manages users and system

b. List of use-cases

1. **Login** – User enters credentials and accesses the system.
2. **Logout** – User exits the system securely.

3. **Add Contact** – User adds new contact details.
4. **Edit Contact** – User updates existing contact details.
5. **Delete Contact** – User removes a contact.
6. **View Contacts** – User views all contacts.
7. **Search Contacts** – User searches contacts using keywords.
8. **Manage Users** – Admin manages system users.

c. Use case diagram

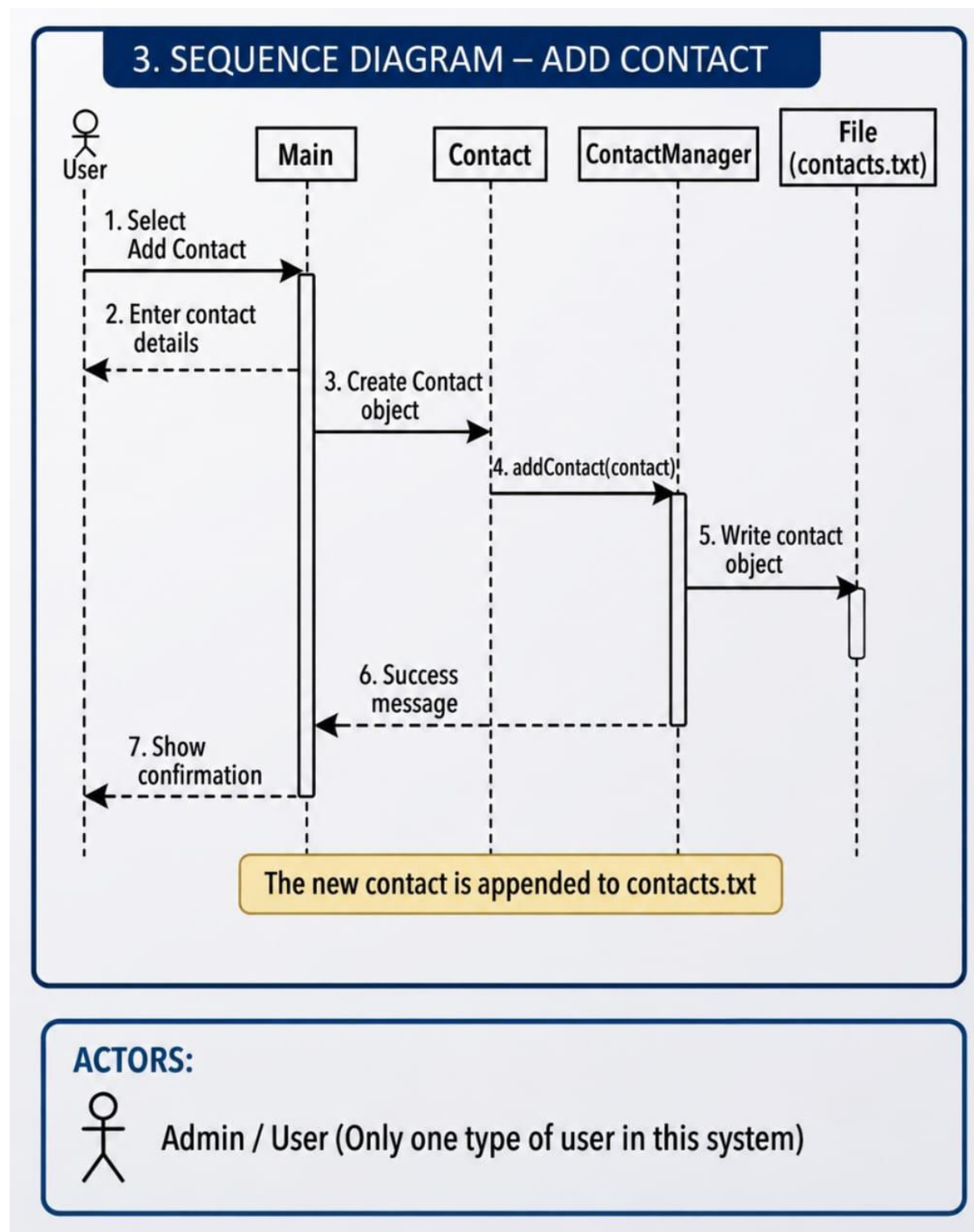


4. Sequence diagram (use UML notations)

a. Use Case: Add Contact

Scenario:

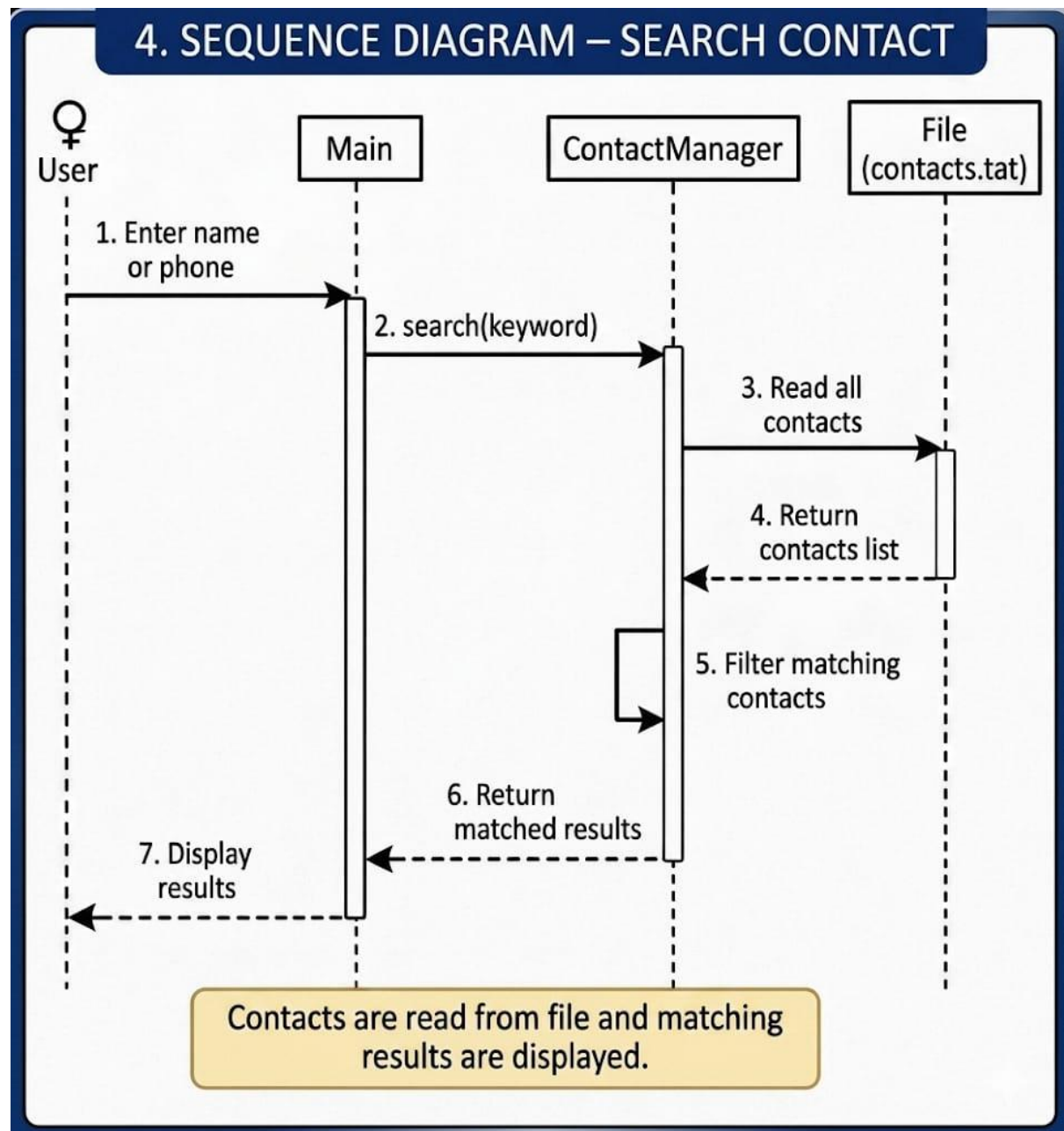
The user logs into the system and enters contact details. The system validates the input, stores it in the database, and displays a confirmation message.



b. Use Case: Search Contact

Scenario:

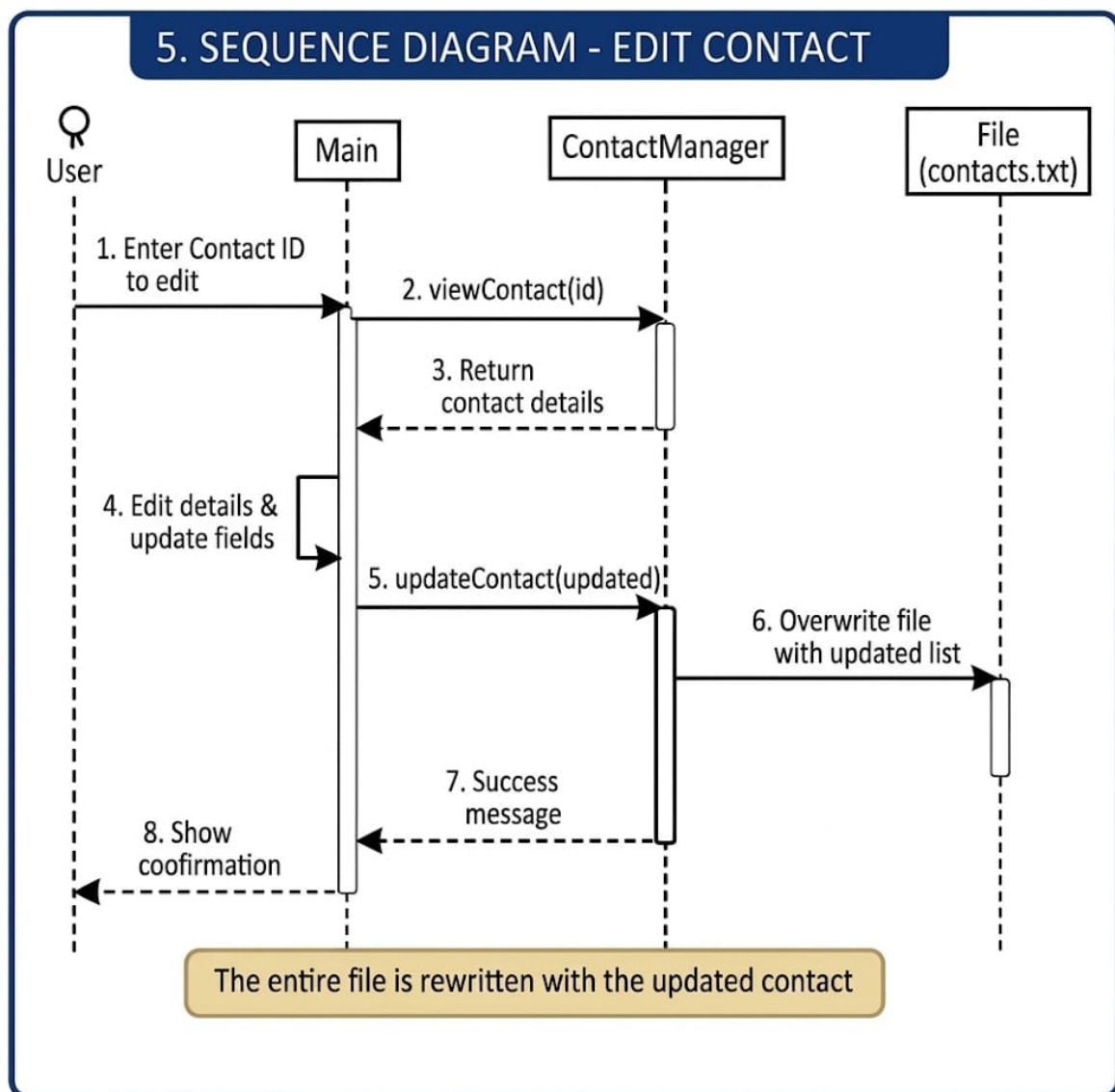
The user enters a keyword to search for a contact. The system processes the request, retrieves matching contacts from the database, and displays the results.



c) Use Case - Edit Contact

Scenario:

The user selects Edit Contact and enters the contact ID. The system retrieves and displays the contact details. The user updates the required fields. The system saves the updated contact and shows a confirmation message.



d) Use Case: Delete Contact

Scenario:

The user selects Delete Contact and enters the contact ID. The system finds and removes the contact from storage. The updated data is saved, and a confirmation message is displayed.

